Notes on BRST Quantization

1. Gauge Redundancy in Yang–Mills Theory

Consider the Yang–Mills action

$$S_{YM} = -\frac{1}{4}\int d^4x\, F^a_{\mu\nu}F^{a\mu\nu}.$$

The field strength is

$$F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + gf^{abc}A^b_\mu A^c_\nu.$$

The theory possesses a local gauge symmetry.

Introduce $A_\mu = A^a_\mu T^a$, with Lie algebra $[T^a, T^b] = if^{abc}T^c$. A finite gauge transformation acts as

$$A_\mu \to A^U_\mu = UA_\mu U^{-1} - \frac{i}{g}(\partial_\mu U)U^{-1}.$$

For an infinitesimal transformation, $U(x) = 1 + i\alpha^a(x)T^a$, we obtain

$$\delta A^a_\mu = D^{ab}_\mu \alpha^b,$$

where $D^{ab}_\mu = \delta^{ab}\,\partial_\mu + gf^{acb}A^c_\mu$ is the adjoint covariant derivative.

Thus many different field configurations correspond to the same physical state. The space of gauge fields therefore contains redundancy $A_\mu \sim A^U_\mu$.

2. The Faddeev–Popov Procedure

Choose a gauge-fixing function $G^a(A) = 0$. (For example, Lorenz gauge $G^a(A) = \partial^\mu A^a_\mu$.)

We want the path integral to intersect each gauge orbit exactly once. Insert the identity

$$1 = \Delta_{FP}[A]\int \mathcal{D}\alpha\delta\big(G(A^\alpha)\big),$$

where $\Delta_{FP}[A]$ is the Faddeev–Popov determinant.

Substituting this into the path integral gives

$$Z = \int \mathcal{D}A\Delta_{FP}[A]\delta\big(G(A)\big)e^{iS[A]}.$$

Under an infinitesimal gauge transformation $\delta A^a_\mu = D^{ab}_\mu \alpha^b$,

$$G^a(A^\alpha) = G^a(A) + \int d^4y \frac{\delta G^a(A(x))}{\delta A^b_\mu(y)} D^{bc}_\mu(y)\alpha^c(y).$$

Define the operator

$$M^{ac}(x,y) = \frac{\delta G^a(A^\alpha(x))}{\delta \alpha^c(y)},$$

then

$$\Delta_{FP}[A] = detM.$$

The determinant may be represented using Grassmann variables.

For fermionic variables,

$$\int d\bar{c}dce^{-\bar{c}Mc} = detM.$$

Introduce anticommuting scalar fields: ghost $c^a(x)$, and anti-ghost $\bar{c}^a(x)$, then

$$\Delta_{FP} = \int \mathcal{D}c\mathcal{D}\bar{c}\, exp\big[i\int d^4x\bar{c}^a\,\partial^\mu D^{ab}_\mu c^b\big].$$

The complete action becomes $S = S_{YM} + S_{gf} + S_{ghost}$, w/

$$S_{ghost} = \int d^4x \bar{c}^a \, \partial^\mu D_\mu^{ab} c^b.$$

(Consider finite-dim matrix M_{ij}, introduce Grassmann varibles $c_i, \bar{c}_i$ that satisfies

$$c_i c_j = -c_j c_i.$$

Consider 1 dim

$$I = \int d\bar{c}dce^{-\bar{c}Mc} = \int d\bar{c}dc(1 - \bar{c}Mc),$$

Using $\int dc = 0, \int dcc = 1$,only $\bar{c}c$ term contributes $I = -M \int d\bar{c}dc\bar{c}c = M$, thus

$$\int d\bar{c}dce^{-\bar{c}Mc} = M$$

Similar for n dim, one gives $I = \det M$

For field theory, it becomes $\int d^4x d^4y \bar{c}^a(x) M^{ab}(x,y) c^b(y)$, thus

$$detM = \int \mathcal{D}c\mathcal{D}\bar{c}e^i \int d^4\, x d^4y \bar{c}^a(x) M^{ab}(x,y) c^b(y)$$

For Lorenz gauge

$$M^{ab}(x,y) = \partial^\mu D_\mu^{ab} \delta(x-y),$$

Thus

$$\Delta_{FP} = \int \mathcal{D}c\mathcal{D}\bar{c} \exp\left[i \int d^4x \bar{c}^a \, \partial^\mu D_\mu^{ab} c^b\right]$$

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3. BRST Symmetry

Gauge fixing appears to destroy gauge invariance. Nevertheless, the quantum theory must still remove unphysical degrees of freedom consistently. The original gauge symmetry survives in a hidden fermionic form. This residual symmetry is BRST symmetry. Conceptually: BRST symmetry is obtained by replacing the infinitesimal gauge parameter with a Grassmann ghost field.

Start from Lorenz gauge $\delta A_\mu^a = D_\mu^{ab} \alpha^b$, define the BRST transformation $\delta_B A_\mu^a = \lambda D_\mu^{ab} c^b$, where λ is a Grassmann-odd constant.

Conventionally one introduces the BRST differential s

$$\boldsymbol{sA_\mu^a = D_\mu^{ab} c^b}.$$

The ghost transformation cannot vanish. The reason is that gauge transformations themselves satisfy a non-Abelian Lie algebra$[\delta_{\alpha_1}, \delta_{\alpha_2}] = \delta_{[\alpha_1, \alpha_2]}$. BRST symmetry must encode this algebraic structure. The correct transformation is

$$\boldsymbol{sc^a = -\frac{1}{2} g f^{abc} c^b c^c}.$$

This equation is essentially the Maurer–Cartan structure equation of the gauge group.

Introduce an auxiliary Nakanishi–Lautrup field B^a. Define $\boldsymbol{s\bar{c}^a = B^a, sB^a = 0}$.

The defining property of BRST symmetry is $s^2 = 0$. This is analogous to the exterior derivative in differential geometry $d^2 = 0$. BRST cohomology is built entirely on this property.

4. BRST-exact gauge fixing

For Lorenz gauge, we want the quantized action to include gauge-fixing term $\mathcal{L}_{gf} = -\frac{1}{2\xi}(\partial^\mu A_\mu^a)^2$ and ghost term $\mathcal{L}_{gh} = -\bar{c}^a\,\partial^\mu D_\mu^{ab} c^b$.

The gauge-fixing and ghost terms can be written as a BRST-exact expression

$$S_{gf+gh} = s\Psi,$$

where Ψ is the gauge-fixing fermion.

Using Nakanishi–Lautrup field B^a, one is able to find

$$\mathcal{L}_{gf} = B^a\,\partial^\mu A_\mu^a + \frac{\xi}{2} B^a B^a.$$

We want $s\Psi$ to generate $B^a\,\partial^\mu A_\mu^a$, $B^a B^a$, and $\bar{c}^a\,\partial^\mu D_\mu^{ab} c^b$.

Note that

$$s(\bar{c}^a\,\partial^\mu A_\mu^a) = B^a\,\partial^\mu A_\mu^a - \bar{c}^a\,\partial^\mu D_\mu^a c^b,$$

and

$$s\left(\frac{\xi}{2}\bar{c}^a B^a\right) = \frac{\xi}{2} B^a B^a.$$

So define

$$\Psi = \int d^4x \bar{c}^a \left(\partial^\mu A_\mu^a + \frac{\xi}{2} B^a\right).$$

Give

$$S_{gf+gh} = s\Psi = \int d^4x \left[B^a\left(\partial^\mu A_\mu^a + \frac{\xi}{2} B^a\right) - \bar{c}^a\,\partial^\mu D_\mu^{ab} c^b\right].$$

The full quantum action is therefore BRST invariant: $sS = 0$.

5. Current and charge

Suppose a set of fields ϕ_i transforms under a continuous fermionic symmetry as

$$\delta\phi_i = \varepsilon s\phi_i.$$

If the Lagrangian varies by a total derivative,

$$\delta\mathcal{L} = \varepsilon\,\partial_\mu K^\mu,$$

then Noether's theorem gives the conserved current

$$j^\mu = \sum_i \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_i)} s\phi_i - K^\mu.$$

The conserved charge is

$$Q = \int d^3x\, j^0.$$

To construct the Noether current explicitly, compute the canonical conjugate momenta. For the gauge field,

$$\Pi^{a\mu\nu} = \frac{\partial\mathcal{L}}{\partial(\partial_\mu A_\nu^a)}.$$

Only the Yang–Mills term contributes

$$\Pi^{a\mu\nu} = -F^{a\mu\nu}.$$

For the ghost fields,

$$\frac{\partial \mathcal{L}}{\partial(\partial_\mu c^a)} = -\partial^\mu \bar{c}^a,$$

and

$$\frac{\partial \mathcal{L}}{\partial(\partial_\mu \bar{c}^a)} = -D^{\mu ab} c^b.$$

The field B^ahas no derivatives and therefore no canonical momentum.

Gauge field contribution gives

$$j^\mu_{(A)} = -F^{a\mu\nu} D^{ab}_\nu c^b.$$

Ghost contribution

$$j^\mu_0 = (-\partial^\mu \bar{c}^a)\left(-\frac{1}{2} g f^{abc} c^b c^c\right) = \frac{1}{2} g f^{abc} (\partial^\mu \bar{c}^a) c^b c^c.$$

Anti-ghost contribution

$$j^\mu_{(\bar{c})} = -B^a D^{\mu ab} c^b.$$

Taken together

$$j^\mu_B = -F^{a\mu\nu} D^{ab}_\nu c^b + \frac{1}{2} g f^{abc} (\partial^\mu \bar{c}^a) c^b c^c + B^a D^{\mu ab} c^b$$

w/ $\partial_\mu j^\mu_B = 0$.

BRST charge

$$Q_B = \int d^3x [-F^{a0\nu} D^{ab}_\nu c^b + \frac{1}{2} g f^{abc} (\partial^0 \bar{c}^a) c^b c^c + B^a D^{0ab} c^b].$$

Note that BRST charge generates transformation, for any field Φ

$$s\Phi = [iQ_B, \Phi\}_\pm,$$

or say

$$[iQ_B, A^a_\mu] = D^{ab}_\mu c^b, \{Q_B, c^a\} = -\frac{1}{2} g f^{abc} c^b c^c.$$

Note that, classical $s^2 = 0$, while quamtum becomes $Q_B^2 = 0$ or $\{Q_B, Q_B\} = 0$.

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